

Kiyan Tavangar

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Education

PhD Candidate at Columbia University

September 2022 - present

Advisors: Kathryn V. Johnston, Melissa Ness, Adrian Price-Whelan

Awards: National Osterbrock Leadership Program Fellow (2023-2025), Lead Teaching Fellow (2024-2025)

Provost Scholar at the University of Chicago, Chicago, IL

Graduated: 06/21

Astrophysics B.S. (Honors, Cum Laude), Mathematics B.S.

Research Interests:

Dark matter, Local Group, near-field cosmology, stellar streams, galactic dynamics, disk dynamics, stellar populations

Research Experience

Research with Prof. Kathryn Johnston at Columbia University

September 2022 - present

Part of PhD thesis: exploring disk dynamics by focusing on the formation and evolution of vertical phase-space spirals. I analyze the changes in phase-spiral morphology across space and time in simulations and data.

Research with Dr. Adrian Price-Whelan at Columbia University and Flatiron Institute

2021 - present

Part of PhD thesis: Building an efficient and accessible stellar stream density modeling framework which improves on previous methods by being able to characterize the stream features most likely to constrain dark matter.

Research with Prof. Melissa Ness at Columbia University

September 2023 - October 2024

Understanding halo dissolution in the Milky Way using empirical methods. The objective was to combine chemical and dynamical methods to quantify how much substructure is present in the halo.

Research with Prof. Alex Drlica-Wagner at the University of Chicago

January 2019 - June 2021

I searched for previously undetected ultra-faint dwarf galaxies and stellar streams in new data. I then analyzed the Phoenix stellar stream by creating a density model and found unexpected small scale density deviations.

Research with Prof. Michael Gladders at the University of Chicago

January 2020 - June 2021

Research with the COOL-LAMPS collaboration focused on the detection of new strong gravitational lenses. I helped publish an analysis on the brightest redshift 5 galaxy in the universe to that point.

Teaching Experience

Co-instructor for *Physics for Poets*

Spring 2026

I am co-teaching the *Physics for Poets* course with Professor Jeremy Dodd at Taconic State Correctional Facility. This is a Columbia course that incarcerated students are taking for credit. Professor Dodd and I split the work for the course approximately equally.

Instructor of *Introduction to Research Skills in Astrophysics*

Spring 2024

I developed and am teaching a course to help undergraduates interested in astronomy get started with research. The course is designed for undergraduates with no research experience. This course is heavily influenced by the success of the StreamTeamTNG program described below.

Organizer and Instructor for the STEM Prison Outreach Program**Spring 2024 - present**

I founded a STEM outreach program at MDC Brooklyn and continue to be the lead organizer. We have biweekly events led by a rotating cast of Columbia scientists who present an overview of their field. In addition to recruiting volunteers, I also generally lead the first session of each semester with an introduction to astronomy.

Lead Teaching Fellow at Columbia University**August 2024 - June 2025**

The Lead Teaching Fellows design and run pedagogical workshops twice a year, among other responsibilities. I ran workshops on how to assign work in the age of AI and on mentoring in STEM.

Mentor in StreamTeamTNG program**January 2023 - August 2023**

StreamTeamTNG was an extracurricular program designed to teach undergraduates interested in astronomy how to do research. It used example projects in stellar streams to teach the basics of working with simulations and data, reading papers, and making presentations. I designed most assignments and led some group meetings.

Teaching Assistant for undergraduate *Stars and Atoms* course**Spring 2023**

Course for non-majors of all years. I hosted office hours weekly and helped design the final project.

Teaching Assistant for undergraduate *Galaxies* course**Fall 2022**

Course for Astronomy or Astrophysics majors (usually third years). I hosted office hours weekly

Selected Publications

Listed in reverse chronological order. Only includes papers to which I contributed significantly.

See <https://ui.adsabs.harvard.edu/user/libraries/S4XWaWaHTr6yn9RvZjuXsQ> for a complete list of papers on which I am a co-author.

1. **Tavangar** et al. (submitted). *Phase-spirals Across Galactic Disks: Using large-scale "macro-spirals" in phase-spiral amplitude to derive perturbation times.*
2. **Tavangar** et al. (2026). Phase spirals across galactic disks I: Exploring dynamical influences on winding. *The Astrophysical Journal*, Volume 1005(1) 120. <https://ui.adsabs.harvard.edu/abs/2026ApJ..1005..120T/abstract>
3. Widmark, **Tavangar** et al. (2026). The phase spiral's origin and evolution: indications from its varying properties across the Milky Way disk. *The Astrophysical Journal*, Volume 999(1), 52. <https://ui.adsabs.harvard.edu/abs/2026ApJ...999...52W/abstract>
4. Chiti & **Tavangar** et al. (2025). DELVE-ing into the Milky Way's Globular Clusters: Assessing extra-tidal features in NGC 5897, NGC 7492, and testing detectability with deeper photometry. *The Astronomical Journal*, Volume 170(5), 294. <https://ui.adsabs.harvard.edu/abs/2025arXiv250711629C/abstract>
5. **Tavangar** and Price-Whelan (2025). Inferring the Density and Membership of Stellar Streams with Flexible Models: The GD-1 Stream in Gaia Data Release 3. *The Astrophysical Journal*, Volume 988(1), 45. <https://ui.adsabs.harvard.edu/abs/2025ApJ...988...45T/abstract>
6. **Tavangar** et al. (2022). From the Fire: A Deeper Look at the Phoenix Stream. *The Astrophysical Journal* Volume 925(2), 118. <https://ui.adsabs.harvard.edu/abs/2022ApJ...925..118T/abstract>
7. Shipp, Price-Whelan, **Tavangar**, et al. (2020). Discovery of Extended Tidal Tails Around the Globular Cluster Palomar 13. *The Astronomical Journal*, Volume 160(5), 244. <https://ui.adsabs.harvard.edu/abs/2020AJ....160..244S/abstract>

Presentations

Listed in reverse chronological order. Invited contributions are asterisked.

1. **Tavangar, K.** et al. (2026). *Timing the Phase Spirals' Origin Using Large-Scale Correlations in Their Morphologies*. Poster presented at "HOW STARS MOVE AROUND AND WHAT THEY TELL US ABOUT GALAXIES' MASSES, FORMATION AND EVOLUTION – From the Milky Way to Giant Elliptical Galaxies", Sexten, Alta Pusteria, Italy.
2. **Tavangar, K.** (2026). *Exploring dynamical influences on phase spiral winding across galactic disks in simulations*. Contributed talk at "Galactic Bars, Bulges and Disks" (AstroLipari 2026), Lipari Island, Italy.
3. **Tavangar, K.** (2026). *Using Phase Spirals to Understand the Milky Way – Sgr Merger*. Talk at the Hernquist Group Meeting at the Center for Astrophysics. Cambridge, Massachusetts, USA.
4. **Tavangar, K.** (2025). *Correlating phase spirals and macro-snails using BFEs and mSSA*. Contributed talk at "Winding, Unwinding and Rewinding the Gaia Phase Spiral" at the Lorentz Conference Center. Leiden, The Netherlands.
5. **Tavangar, K.** (2025). *Inferring the density and membership of stellar streams: Application to GD-1*. Contributed talk at "Gravity in the Local Group" at Carnegie Mellon University. Pittsburgh, PA, USA.
6. **Tavangar, K.** (2025). *Macro and Micro Snails from a Single Satellite Flyby with EXP*. Contributed talk at "AAS245" at the Gaylord National Resort and Convention Center. National Harbor, MD, USA.
7. **Tavangar, K.** (2024). *Macro and Micro Snails from a Single Satellite Flyby with EXP*. Contributed talk at "Waves in the Milky Way Disk" conference at Shanghai Jiao Tong University (participated remotely). Shanghai, China.
8. ***Tavangar, K** and Mateu, C. (2024). *Galstreams and CATS Tutorial*. Invited tutorial presentation at "Streams24: The Theory Edition" at Durham University. Durham, UK.
9. ***Tavangar, K.** (2024). *Galactic Cannibalism: Milky Way Formation and Hints Towards Dark Matter*. Invited talk for the Columbia chapter of the Society of Physics Students. New York, NY, USA.
10. **Tavangar, K.** (2024). *Stellar Streams: Evidence of Galactic Cannibalism*. Presentation at PlaceExchange as part of NOLP fellows trip. New York, NY, USA.
11. **Tavangar, K.** (2022). *Modeling the Spur in the GD-1 Stream: A way to test the dark matter sub halo population*. Talk at the Survey Science Group meeting at the University of Chicago. Chicago, IL, USA.
12. **Tavangar, K.** (2021). *Stream Modeling of ATLAS and Phoenix in DES*. Contributed talk at the virtual conference "Streams21: Constraints on Dark Matter".

Scientific Collaborations

EXP Collaboration Student Member	2022-present
Community Atlas of Tidal Streams (CATS)	2022-present
Center for Computational Astrophysics (CCA) Dynamics Group at the Flatiron Institute	2021 - 2022

ChicagO Optically selected strong Lenses – Located At the Margins of Public Surveys Collaboration (COOL-LAMPS) Member	2020 – present
Southern Stellar Stream Spectroscopic Survey (S ⁵) Member	2020 – 2022
Dark Energy Survey (DES) Collaboration External Collaborator	2019 – present
DECam Local Volume Exploration Survey (DELVE) Student Member	2019 – present
Kavli Institute of Cosmological Physics Survey Science Group Member	2019 – 2021

Technical Proficiencies

- Experience writing open-source software
- Extensive knowledge of Python
- Working knowledge of MATLAB, Stan, and Scheme
- Experience in lensed source modeling
- Extensive experience in dynamical modeling
- Extensive experience using large datasets
- Three nights remote observing experience with the Blanco 4m Telescope for the DELVE survey